2110201: COMP ENG MATH (2024/2) Quiz 1

Name: Student ID: Entry Number:

1. (1.5 points, 0.5 points each) Answer **True** or **False** for each sub-question in the table provided.

1.1	**1.2**	**1.3**

1.1. $G(a, b, c) = (b - c,\ a + 2)$ is a linear transformation.

1.2. Given matrix $\mathbf{T} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $\mathbf{x} = (1,\ 1)$, and $\mathbf{b} = \mathbf{Tx} = (3,\ 7)$. We can conclude that $(3,\ 7)$ is an image of $(1,\ 1)$ respect to $\mathbf{T}$.

1.3. The vector $(1,\ 1)$ is <u>**not**</u> an eigenvector of the matrix $\begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix}$.

2. (8.5 points) Given a function f such that

$$f(a, b, c, d) = (a + 2b - c,\ b + 3c - 4d,\ -a + c - 2d)$$

Let $\mathbf{A}$ be a matrix that represents the linear transformation f.

2.1. (0.5 points) If the transformation f has a domain of $\mathbf{R}^m$ and a range of $\mathbf{R}^n$ (f: $\mathbf{R}^m \Rightarrow \mathbf{R}^n$) where $\mathbf{m}$ and $\mathbf{n}$ are positive integers. Find $\mathbf{m}$ and $\mathbf{n}$.

2.2. (2 points) Find the matrix $\mathbf{A}$.

2.3. (3 points) Find RRE of the matrix $\mathbf{A}$.

2.4. (2 points) Find all vectors $\mathbf{x}$ such that $\mathbf{Ax} = \mathbf{0}$

2.5. (0.5 points) Is the function f one-to-one? (Answer **Yes**/**No** only) _____

2.6. (0.5 points) Is the function f onto? (Answer **Yes**/**No** only) _____